Lecture 10

Corey Powell

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Midterm 1

- Reminder that Midterm 1 will be on Wednesday, October 14 in class.
- Midterm 1 will cover:
  - Chapters 1-5 and 8-12 in the text, and
  - the material in Lectures 1-8.
- The R material covered in lab will not be on Midterm 1 but is fair game for Midterm 2 and the final.

Midterm 1 (continued)

- You can find example questions for the exam from:
  - Sample midterms posted on Canvas
  - Homework questions
  - Additional questions in the text
  - Review questions
  - Questions discussed in lecture
- You will want to focus more on solving problems and less on memorizing the lectures.

Lecture 10

1. Basic Concepts and Notation
2. Bounds on Joint Probabilities
3. Three Fun Probability Problems
4. Example 1
5. The Law of Total Probability
6. Bayes' Rule

Finish Reading Chapter 15 for next time.

Quiz 5 is on Wednesday at the end of lecture.

Basic Concepts and Notation

Discussion Question

- Suppose you deal two cards from a standard 52 card deck.
- Define events $A =$ "First card is a King" and $B =$ "Second card is an Ace"
- What is $P(A)$?
- What is $P(B)$?
- What is $P(B|A)$?
- Explain in words the frequency interpretation for each of the above.
  - Note that the frequency interpretation of $P(B|A)$ does not involve the word "know."
- Compare this to the subjective interpretation.

Independence

- Events A and B are **independent** if and only if $P(A|B) = P(A)$ and $P(B|A) = P(B)$.
  - Does $P(A|B) = P(A)$ imply that $P(B|A) = P(B)$?
  - We could also write the definition as $P(A \text{ and } B) = P(A)P(B)$.
- In other words, the probability of A is the same whether or not B happens, and vice versa.

Examples of Independence

- Repeated tosses of a coin.
- Repeated draws of a card from a standard 52 card deck.
  - You put back (replace) the card after each draw.
  - Would the repeated draws be independent if you didn't replace the card?
- Voting decisions of individuals in different states who don't know each other.
- Other examples?

Mutually Exclusivity

- Events A and B are **mutually exclusive** if and only if $P(A \text{ and } B) = 0$.
- In other words, it is impossible for both A and B to happen.

Examples of Mutual Exclusivity

- Getting an A on an exam and getting a B on the exam.
- Winning and losing a game.
- Other examples?

Watch Out!!

- Both independent and mutually exclusive can be translated into ordinary English as meaning that the two events have nothing to do with each.
- These are very different concepts!
- Suppose that A and B are both independent and mutually exclusive.
  - What does this say about $P(A)$ or $P(B)$?

Calculation Rules

- It is always true that $P(A \text{ and } B) = P(A)P(B|A)$.
- A and B independent $\rightarrow P(A \text{ and } B) = P(A)P(B)$.
- If A and B mutually exclusive, then
  - $P(A \text{ or } B) = P(A) + P(B)$.
- How can we use this fact to show that
  - $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$?
- If A and B are independent, then
  - $P(A \text{ or } B) = P(A) + P(B) - P(A)P(B)$
- We can use these rules to put bounds on the **joint probability** $P(A \text{ and } B)$.

Bounds on Joint Probabilities

Joint Probabilities (Example 1)

- Suppose $P(A) = 0.4$ and $P(B) = 0.5$
  - What is $P(A \text{ and } B)$?
  - What is $P(A \text{ and } B)$ if A and B are independent?
  - What is $P(A \text{ and } B)$ if A and B are mutually exclusive?
  - Are there any other possible values that $P(A \text{ and } B)$ could be?
  - Are there any values that $P(A \text{ and } B)$ ***can't*** be?

Joint Probabilities (Example 2)

- Suppose $P(A) = 0.8$ and $P(B) = 0.5$
  - What is $P(A \text{ and } B)$?
  - What is $P(A \text{ and } B)$ if A and B are independent?
  - What is $P(A \text{ and } B)$ if A and B are mutually exclusive?
  - Are there any other possible values that $P(A \text{ and } B)$ could be?
  - Are there any values that $P(A \text{ and } B)$ ***can't*** be?

Bounds on Joint Probabilities (discussion)

- For any pair of events A and B,
  - What is the maximum possible value of $P(A \text{ and } B)$?
  - What is the minimum possible value of $P(A \text{ and } B)$?
  - What is the maximum possible value of $P(A \text{ or } B)$?
  - What is the minimum possible value of $P(A \text{ or } B)$?

Bounds on Joint Probabilities

- For any pair of events A and B,
  - The maximum possible value of $P(A \text{ and } B)$ is $\min(P(A), P(B))$.
  - The minimum possible value of $P(A \text{ and } B)$ is $\max(P(A) + P(B) - 1, 0)$.
  - The maximum possible value of $P(A \text{ or } B)$ is $\min(P(A) + P(B), 1)$.
  - The minimum possible value of $P(A \text{ or } B)$ is $\max(P(A), P(B))$.

Discussion Exercise

- If A is "rain on Saturday," and B is "rain on Sunday," and $P(A) = 0.4$ and $P(B) = 0.4$,
  - What is the maximum possible value of the probability of rain on both Saturday and Sunday?
  - What is the minimum possible value of the probability of rain on both Saturday and Sunday?

Three Fun Probability Problems

The Birthday Problem

Birthday Problem

- How large must a class be to make the probability of finding at least two people with the same birthday at least 50%?



The Monty Hall Problem

Monty Hall Problem

- Suppose you're on a game show, and you're given the choice of three doors: Behind one door is a car; behind the others, goats. You pick a door, say No. 1, and the host, who knows what's behind the doors, opens another door, say No. 3, which has a goat. He then says to you, "Do you want to pick door No. 2?" Is it to your advantage to switch your choice?



Airplane

Airplane!

- One hundred people line up to board an airplane. Each has a boarding pass with assigned seat. However, the first person to board has lost his boarding pass and takes a random seat. After that, each person takes the assigned seat if it is unoccupied, and one of unoccupied seats at random otherwise. What is the probability that the last person to board gets to sit in his assigned seat?



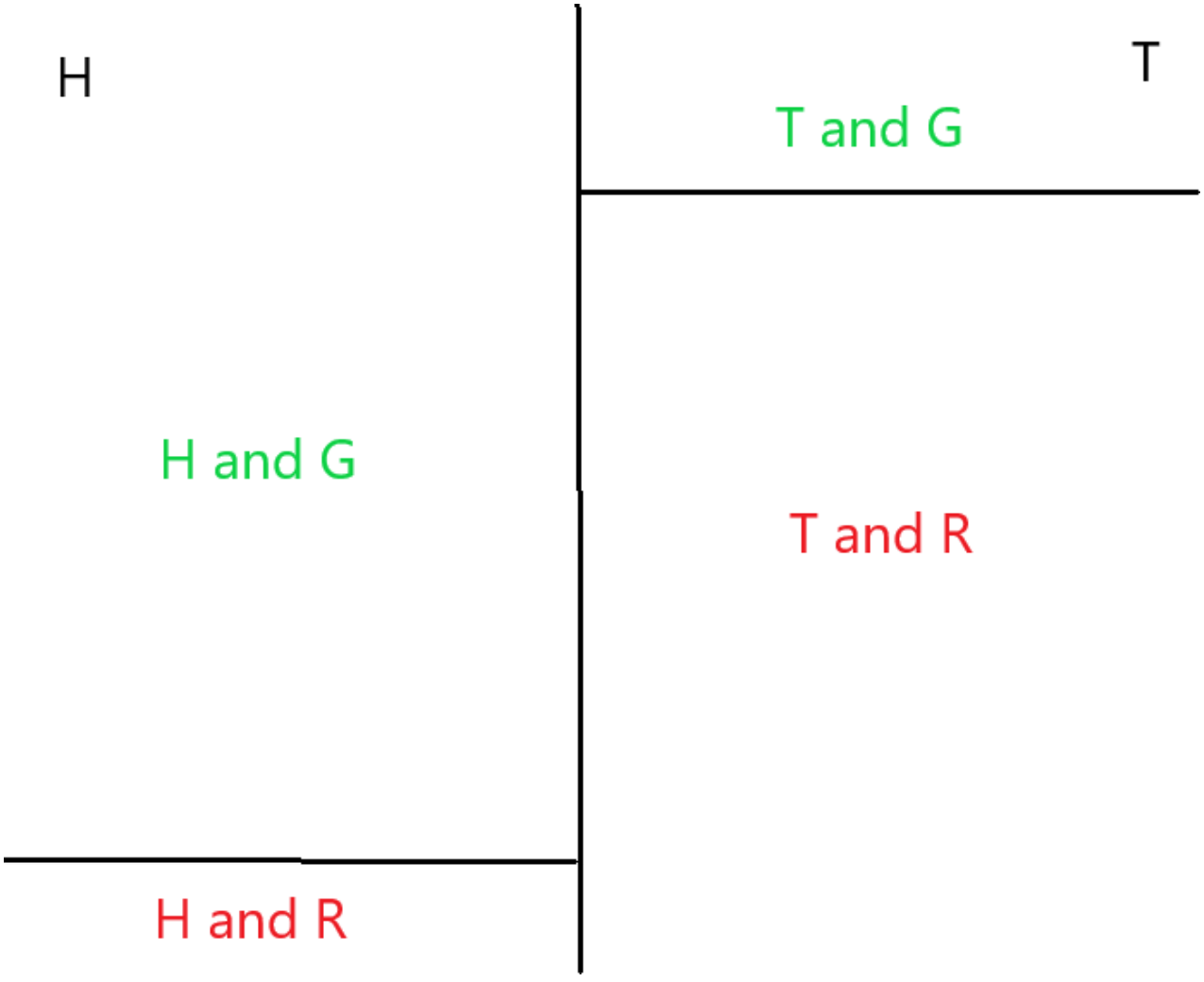
Example 1

- In jar A there is 1 red marble and 9 green marbles.
- In jar B there are 8 red marbles and 2 green marbles.
- You flip a coin.
  - If it lands heads, you pick a marble at random out of jar A.
  - If it lands tails, you pick a marble at random out of jar B.

Example 1 (Continued)

- What is the chance the coin lands heads?
  - We'll call this $P(H)$
- Given the coin lands heads, what is the chance of picking a red marble?
  - We'll call this $P(R|H)$.
- Given that the coin lands tails, what is the chance of picking a red marble?
  - We'll call this $P(R|T)$.
- What is the unconditional chance of picking a red marble?
  - We'll call this $P(R)$.

Venn Diagram



Law of Total Probability

- Note that $R \text{ and } H$ and $R \text{ and } T$ are mutually exclusive, and together they comprise R.
- Hence $P(R) = P(R \text{ and } H) + P(R \text{ and } T)$.
- We can now use the definition of conditional probability to rewrite this as:
  - $P(R) = P(R|H)P(H) + P(R|T)P(T)$.
- This is the **Law of Total Probability**.
- We can now calculate $P(R) = 0.1 * 0.5 + 0.8 * 0.5 = 0.45$.

Example 1 (Continued)

- Given you pick a red marble, what is the chance the coin landed heads? We'll call this $P(H|R)$.
- The last question is more challenging.
- Do you think that the chance is:
  - (i) about 50%,
  - (ii) much less than 50%, or
  - (iii) much more than 50%?
- Why?

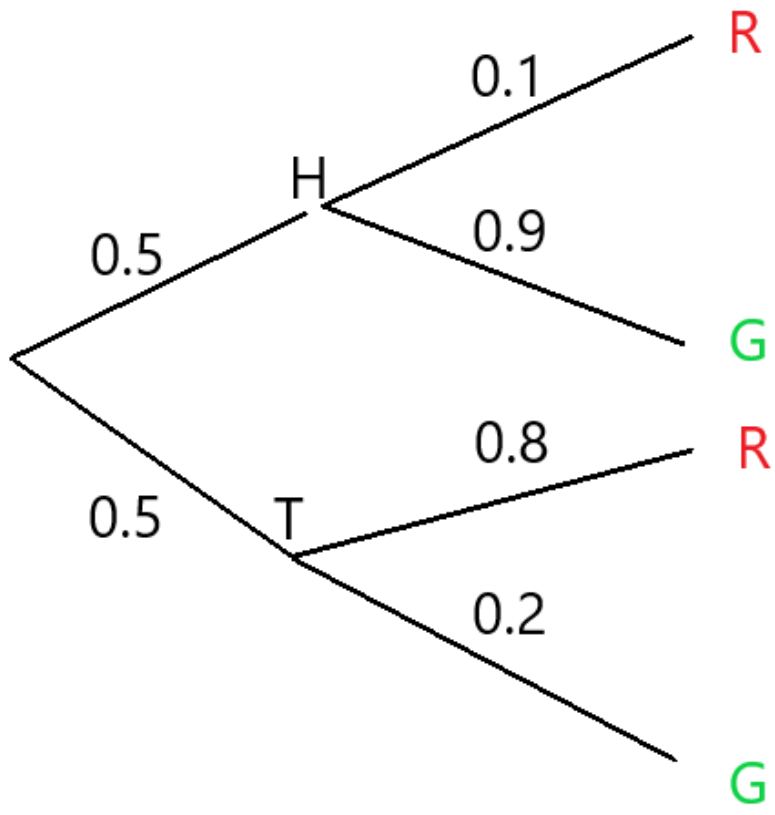
Bayes' Rule

- By the definition of conditional probability, we know that for two events A and B:
  - $P(A|B) = P(A \text{ and } B)/P(B)$.
- We also know from the definition of conditional probability that:
  - $P(A \text{ and } B) = P(B \text{ and } A) = P(B|A) * P(A)$.
- Hence $P(A|B) = P(B|A) * P(A)/P(B)$
- This is **Bayes' Rule**

Bayes' Rule (continued)

- Bayes' Rule allows us to calculate a conditional probability by reversing the direction of conditionality.
  - One direction is frequently more intuitive than the other.
  - The design of Example 1 makes $P(R|H)$ more intuitive than $P(H|R)$.
- We can now calculate $P(H|R) = P(R|H) * P(H)/P(R)$.
- This gives $P(H|R) = 0.1 * 0.5/0.45 = 0.11$.

Bayes' Rule in a Tree Diagram



Bayes' Rule in a Tree Diagram (Continued)

- To find the probability of a path, you multiply the probabilities along the path.
- To find the probability of an event, you add the probabilities of the paths that lead to that event.
- With this method, $P(H|R) = P(H \text{ and } R)/P(R)$
  - There is one path that leads to H and R and there are two paths that lead to R.
  - Hence $P(H|R) = 0.5 * 0.1/(0.5 * 0.1 + 0.5 * 0.8) = 0.11$.

Example 2

- A large company randomly tests its employees for drug use, and fires anyone who tests positive.
- As a result of this policy, only one out of 1000 employees use drugs.
- The test the company uses is not perfect, but if an employee is using drugs, there is a 98% chance of testing positive.
- If an employee is drug-free, there is a 99% chance of testing negative.
- If a randomly chosen employee tests positive for drugs, what is the chance the employee actually uses drugs?

Odds

- If $P(A)$ is the probability of some event A, then the **odds** of A are $P(A)/(1 - P(A))$.
- We can write the odds as $P(A)/P(\sim A)$, where $\sim A$ is the opposite event of A.
- What are the odds of A if $P(A)$ is:
  1. 0?
  2. 0.0001?
  3. 0.25?
  4. 0.5?
  5. 0.8?
  6. 0.99?

Odds (Continued)

- If the probability is small, then the odds and the probability are very close.
- If the probability is 0.5, then the odds are 1.
- The odds tend to ∞ as the probability tends to 1.
- If the odds are o and the probability p, then $p = o/(1 + o)$.
- Gamblers generally refer to the odds as opposed to the probability.

Bayes' Rule in Terms of Odds

- We can use Bayes' Rule to write the odds of $A|B$, referred to as the **posterior odds**, as:

$$P(A|B)/P(\sim A|B) = \frac{P(B|A) * P(A)/P(B)}{P(B|\sim A) * P(\sim A)/P(B)}$$

.

- We can then rearrange the right hand side of this equation as:
  - $(P(A)/P(\sim A)) * (P(B|A)/P(B|\sim A))$.

Bayes' Rule in Terms of Odds (Part 2)

- We call the first term $P(A)/P(\sim A)$ the **prior odds**.
  - This reflects the odds of A before taking into account any other information.
- We call the second term $P(B|A)/P(B|\sim A)$ the **data odds**.
  - This reflects the influence of new data B.

Bayes' Rule in Terms of Odds (Part 3)

- We can now restate Bayes' Rule as:
  $$\text{posterior odds} = \text{prior odds} * \text{data odds}\ .$$
- This gives statisticians a systematic way to update their expectations about the world when new data comes in.
- This is the basis of a branch of statistics called **Bayesian Statistics**
- **Frequentist Statistics**, the approach taken in this course, is the more traditional data analysis approach.

Example 2 Expressed in Terms of Odds

- Prior odds: $(1/1000)/(999/1000) = 1/999$
- Data odds: $0.98/0.01 = 98$
- Posterior odds: $98/999$.
- Probability: $98/(98 + 999) = 98/1,097$.

Discussion

1. Would you describe the drug test as accurate? Explain.
2. If you were a manager at this company, what might you suggest to do differently?

Review Questions

- See Review_questions9-10.pdf and Review_questions11.pdf on Canvas